3rd Year – Second Cycle / Option: CyS
IA for Cybersecurity

Lab 02 : ML/DL techniques in Cybersecurity

Objective

Apply data visualization and analysis techniques to a cybersecurity use case. Each student will select one use case and explore an appropriate cybersecurity dataset using Python or a visualization tool.

Tasks to complete

1. **Choose one cybersecurity use case** from the following:
   - Visualization of network traffic patterns
   - Analysis of log data
   - Visualization of Intrusion Detection alerts
   - Phishing email analysis
   - Password analysis
   - Visualization of data from SIEM systems
   - Visualization of Firewall rules and Network topology
   - Behavioral analysis of malicious software
2. **Select or simulate an appropriate dataset** related to the chosen use case.
3. **Choose a suitable tool** for the analysis:
   a. Python (Pandas, Matplotlib, Seaborn, Plotly, etc.)
   b. Kibana
   c. Another appropriate visualization tool
4. **Explore and analyze the data** and identify relevant patterns, trends, or anomalies.
5. **Inject or identify one anomaly** and clearly indicate where the anomaly occurs.
6. **Produce at least two visualizations** that clearly show the anomaly or relevant cybersecurity pattern.
7. **Define one simple detection rule** to automatically identify the anomaly.
8. **Evaluate the detection rule** on both anomalous and normal data.
9. **Present and explain the main findings.**

Deliverables

- A **well-structured and commented Jupyter Notebook** if Python is used.
- A **report** describing the dataset, methodology, visualizations, detection rule, and main findings.